

Integrating Ground-Based and Satellite Data for Station-Level Air Quality Index (AQI) Prediction in Delhi

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What is AQI?

The Air Quality Index (AQI) is a standardised statistic for measuring and communicating the quality of air in the region in question. It provides a basic, comprehensible picture of how clean or polluted the air is, as well as the health hazards associated with it.

What are the factors that affect the AQI?

Pollutants:

- Particulate matter (PM2.5 and PM10)
- Carbon Monoxide (CO)
- Nitrogen Oxides (NO and NO2)
- Sulfur Dioxide (SO2)
- Ozone (O3)
- Benzene, toluene, xylene, and ammonia (NH3)

City typology:

- Urban
- Rural
- Coastal

Seasonal variations:

- Winter
- Spring
- Summer
- Monsoon
- Autumn

Background and motivation

Air pollution is one of the biggest environmental and public-health challenges in cities. If we want to understand exposure and plan interventions, we need to **predict AQI accurately**, not just report it. Delhi is a stark case: its air quality ranks among the worst globally, and AQI swings **seasonally** and spatially.

Beyond emissions, **vegetation and meteorology** shape dispersion and secondary chemistry. Greener areas can correlate with cleaner air, while stagnant, dry conditions can lock pollution in place. However, many prior studies lean heavily on **ground measurements alone** and rarely integrate **satellite information** like NDVI that captures land-surface context.

Research question and objectives

How well can we explain and predict station-level AQI from pollutants and satellite indicators (NDVI and AOD) in Delhi?

O1: Build a clean, consistent **master dataset** (37 stations, 2017–2024).

O2: Explore spatio-temporal patterns; quantify **correlations** (AQI vs pollutants; NDVI and AOD vs AQI/PM).

O3: Train baseline ML models for AQI regression (and AQI-class classification).

O4 (Next): Compare models across feature sets & transforms.

Our Dataset

Our ground dataset includes daily air quality and pollutant data for 37 stations in Delhi from 2018 to 2024. It covers major pollutants such as PM_{2.5}, PM₁₀, NO, NO₂, SO₂, CO, NH₃, Benzene, Toluene, and Xylene, as well as the calculated Air Quality Index (AQI). This has been made available by DPCC, CPCB, and IMD.

For our satellite data, we're using Sentinel-2 to compute our 6-month means for each station (2018-2023). For AOD, we're using MODIS (MAIAC). This satellite data has been made available by Google Earth Engine.

Platforms

Jupyter Notebook for python and machine learning.

Google Earth Engine for station boundary sketching and satellite data computation.

Data Preprocessing Pipeline

Structure and cleaning

Structure

- For each station:
 - 8 AQI Files containing daily data (2017-2024)
 - 8 pollutant data files containing daily data (2017-2024)
- A total of 592 files for all stations.

Merging into one master CSV file

- Converted the annual AQI data files into a structure that could be merged with the pollutants data.
- Made an AQI file and a pollutants data file for each station. (74 files)

Cleaning and outlier treatment

Cleaning

- Dropped features with >30% missing values.
 - Left with 8 core pollutants.
- Went through the descriptive stats to check for any outliers.

Outlier Treatment

- Winsorised each feature by station at 99.9th percentile.
- Time interpolation upto 3 days; for longer gaps -> monthly mean within stations.
- Deleted rows that no AQI reading.

Feature Engineering and correlation analysis

Feature Engineering

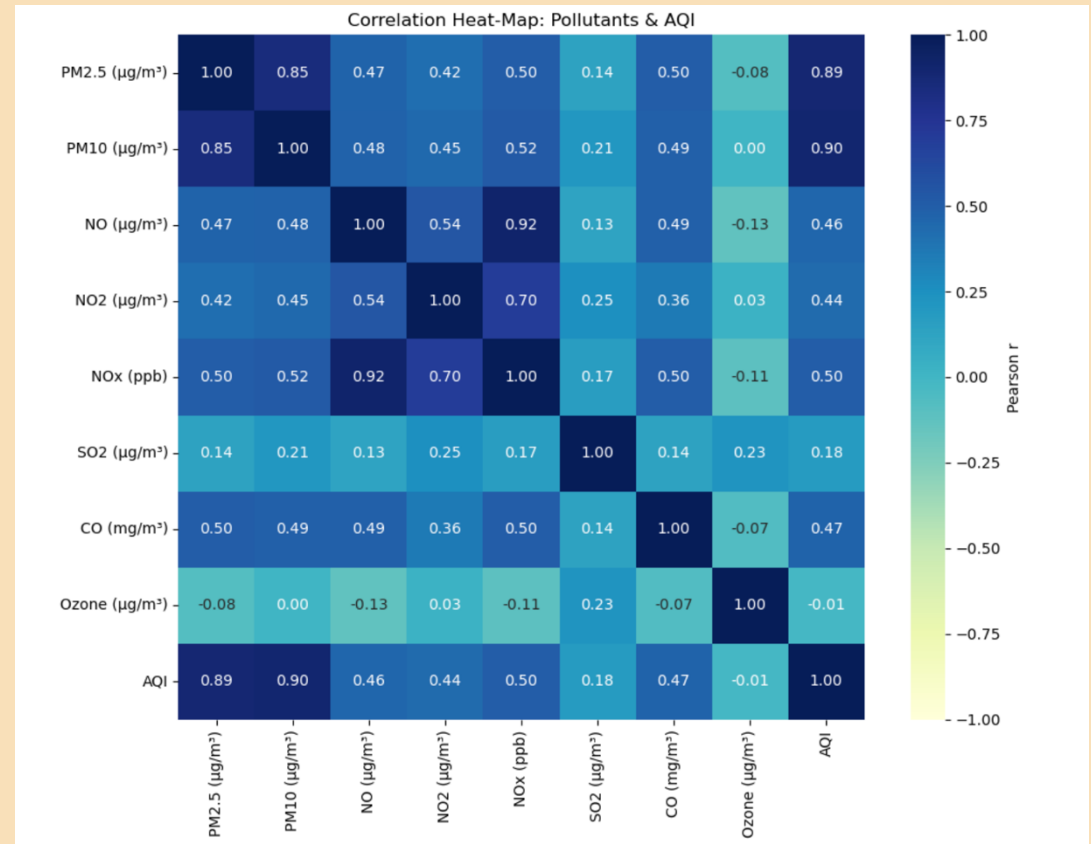
- Created new features:
 - Year, Month and Season.
 - Station_AQI_Mean: Average AQI for each station
 - AQI_Class: Classification as per AQI range (Good to Severe)

Correlation analysis

- Calculated and visualised the correlation between AQI and various pollutants.
- Calculated and visualised the correlation of AQI, PM2.5, PM10 with NDVI.

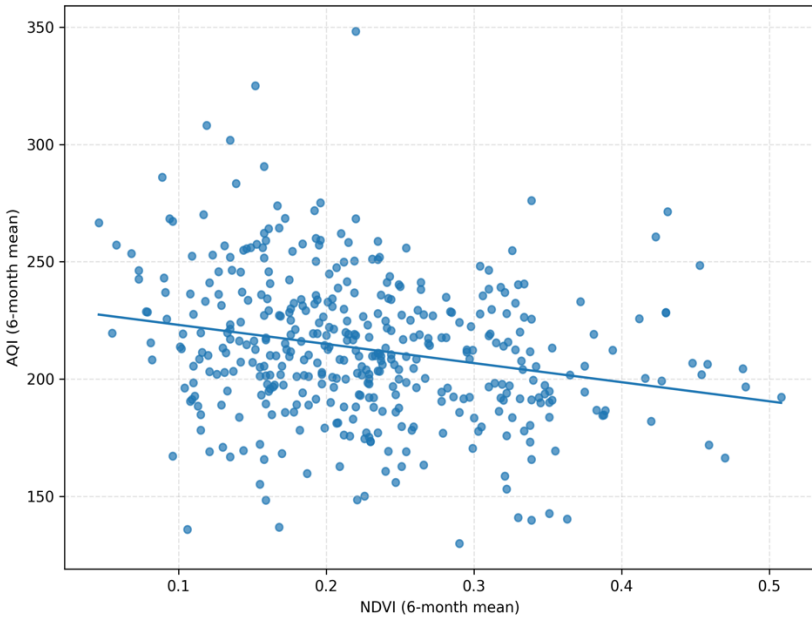
Correlation Analysis (AQI vs Pollutants)

- **PM2.5** and **PM10** show strong correlations with AQI i.e., **0.89** and **0.90** respectively.
- **CO** ($r=0.47$), **NO₂** ($r=0.44$), **NO_x** ($r=0.50$), and **NO** ($r=0.46$) exhibit moderate correlations with AQI.
- **Ozone** shows almost no correlation with AQI.
- **NO**, **NO₂**, and **NO_x** exhibit high inter-correlations (e.g., $r=0.92$ between **NO** and **NO_x**, $r=0.70$ between **NO₂** and **NO_x**). This indicates potential multicollinearity, which could affect regression models. So, we will only use **NO₂** due to its longevity in air.



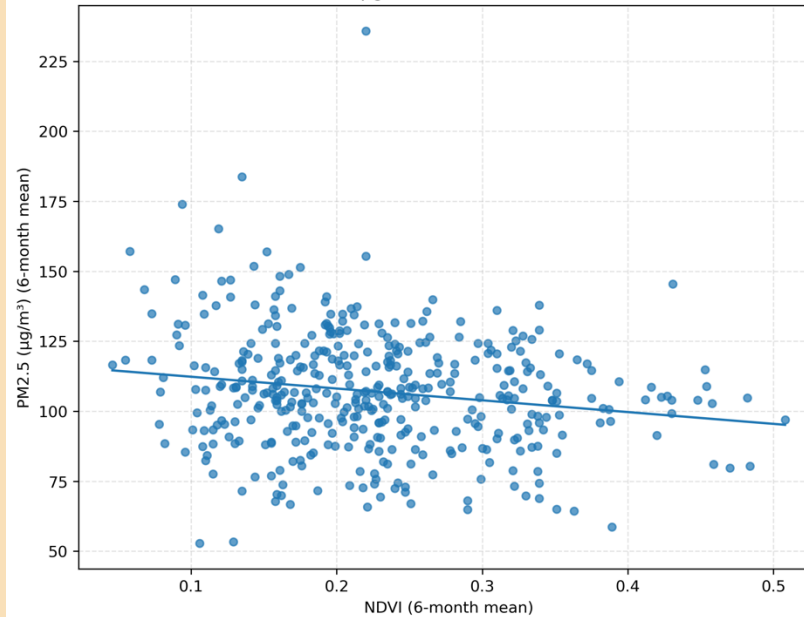
Correlation Analysis (AQI, PM2.5, PM10 vs NDVI)

NDVI vs AQI (All Stations, Half-Year Bins)



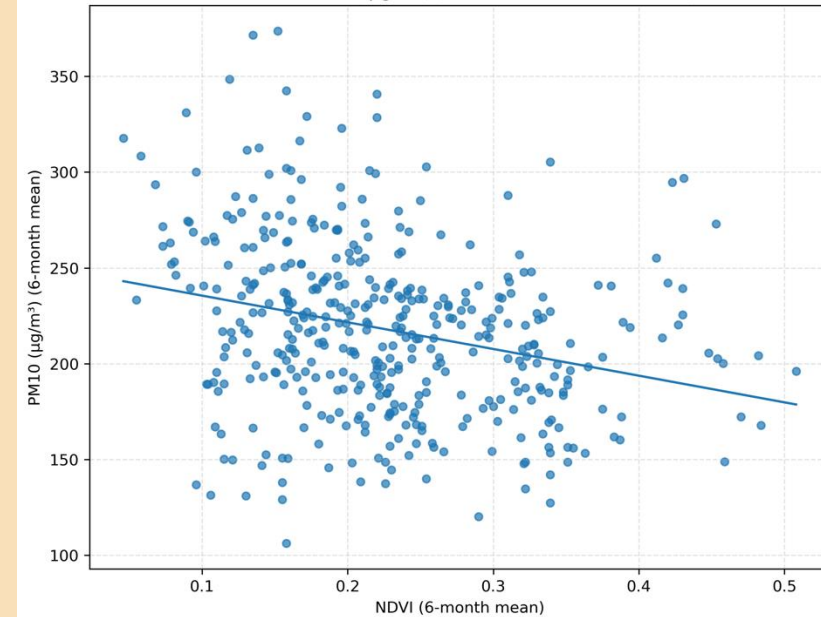
Pearson $r = -0.233$
Spearman $\rho = -0.242$

NDVI vs PM2.5 ($\mu\text{g}/\text{m}^3$) (All Stations, Half-Year Bins)



Pearson $r = -0.174$
Spearman $\rho = -0.155$

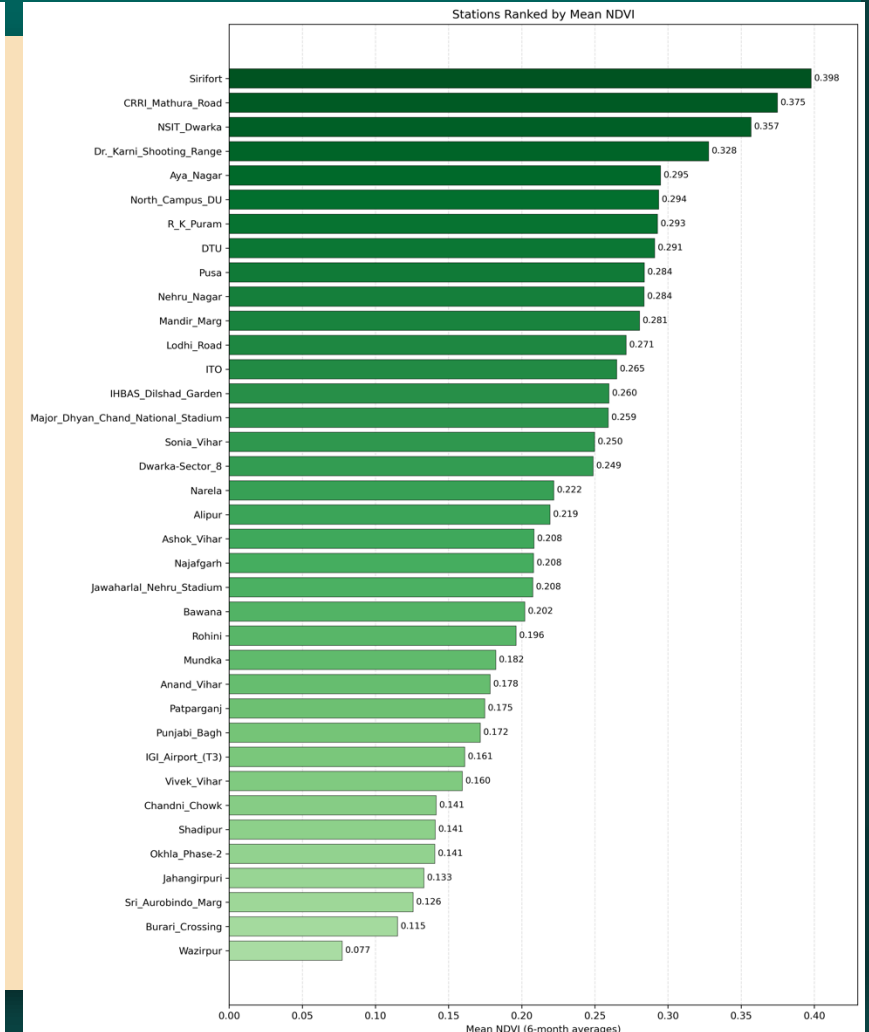
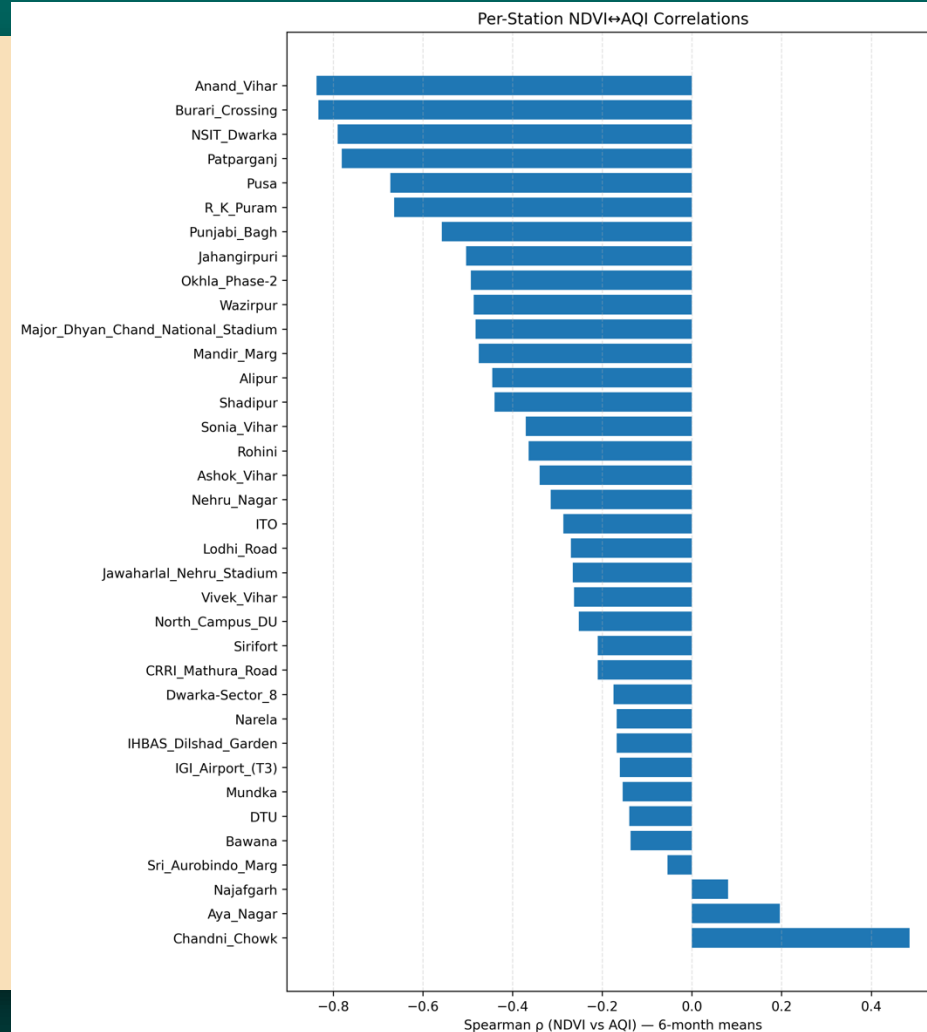
NDVI vs PM10 ($\mu\text{g}/\text{m}^3$) (All Stations, Half-Year Bins)



Pearson $r = -0.270$
Spearman $\rho = -0.292$

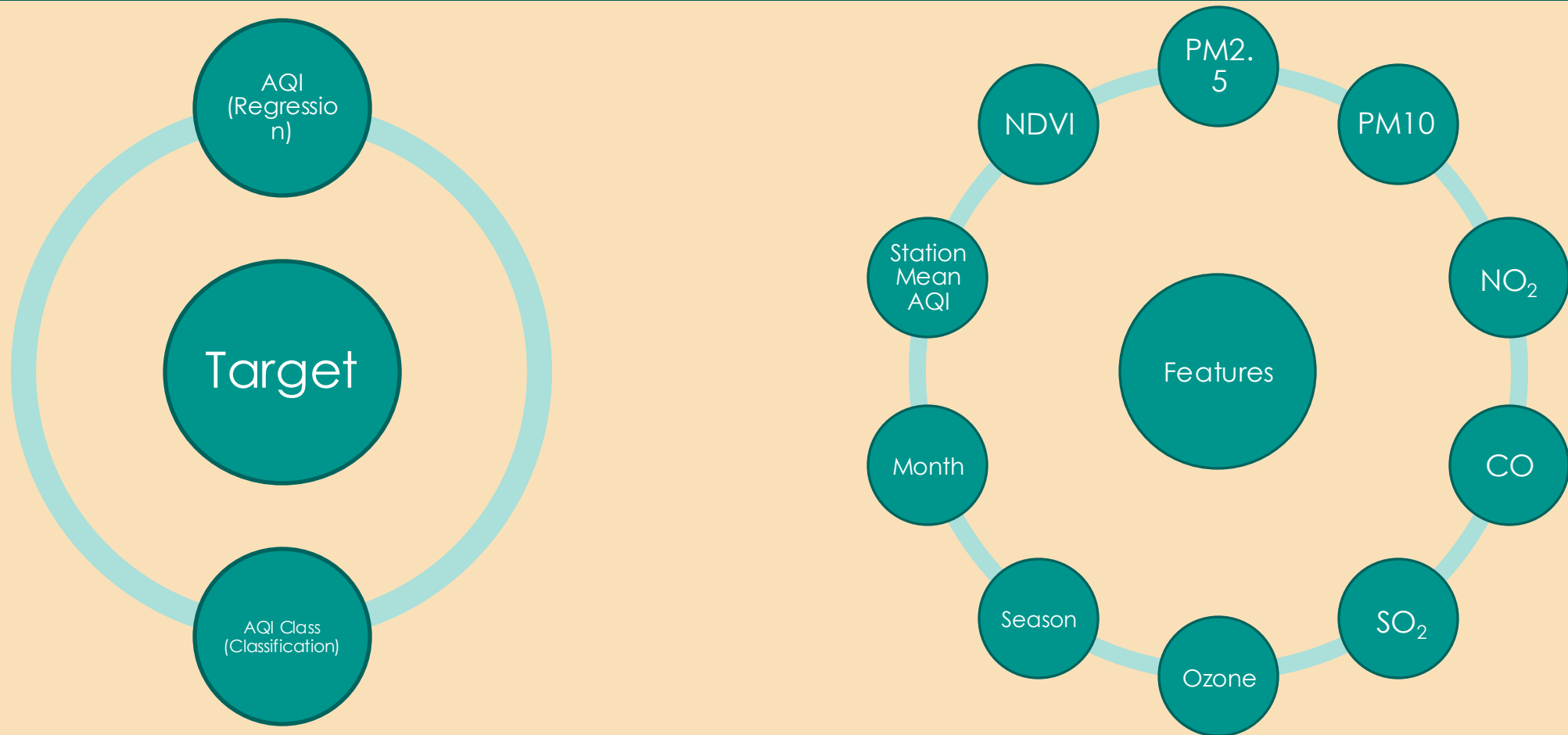
Correlation Analysis (Per station NDVI)

- Strong negatives at **Anand Vihar, Burari Crossing, NSIT Dwarka, Patparganj, and Pusa**.
- Mixed/positive at **Chandni Chowk, Aya Nagar, Najafgarh** (local sources & micro-environment).
- NDVI is a **contextual covariate** (dispersion/land-cover proxy), not a silver bullet.



Target and Feature Set

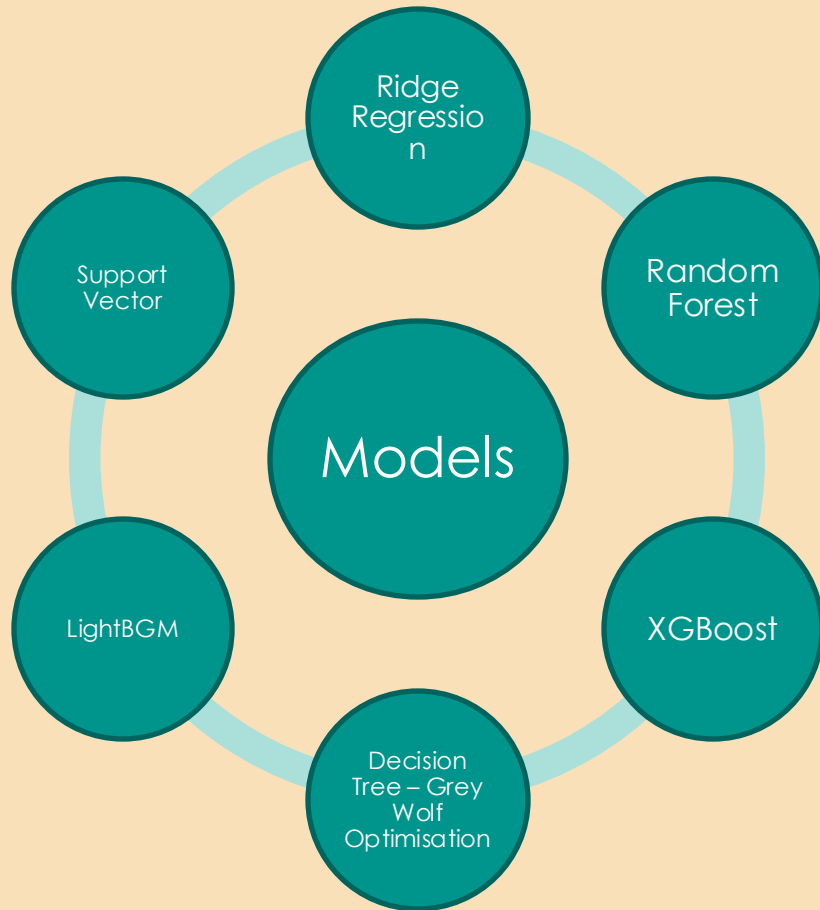
Target and Feature Set





Machine Learning

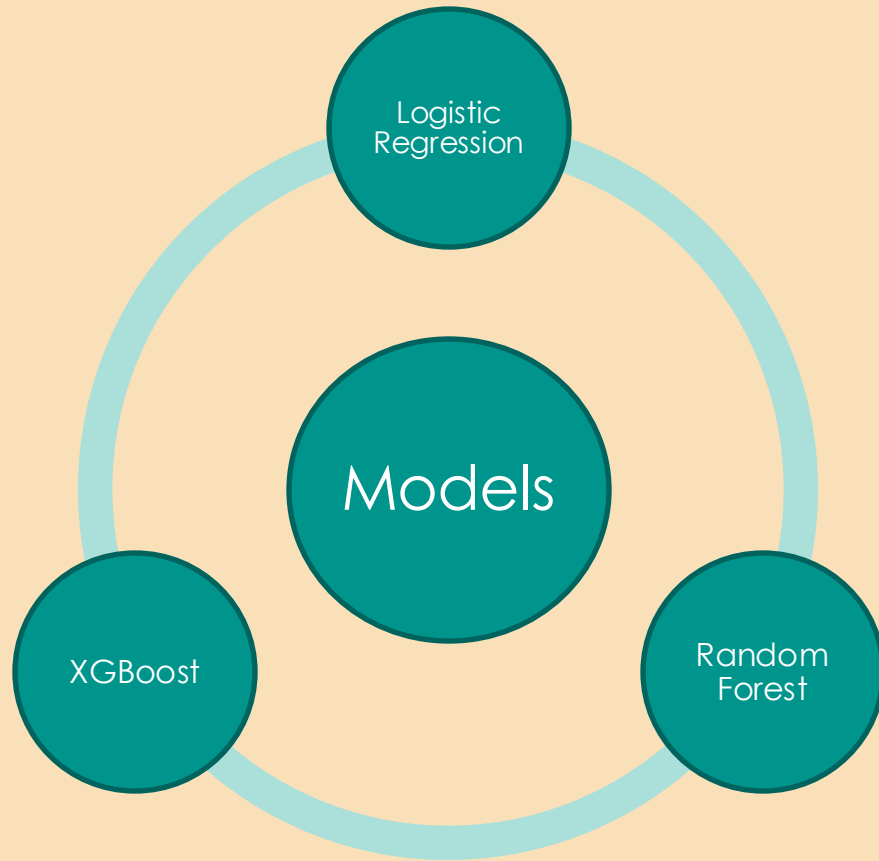
Regression Models



Regression metrics:

- R^2
- RMSE
- MAE
- MSE

Classification Models



Classification metrics:

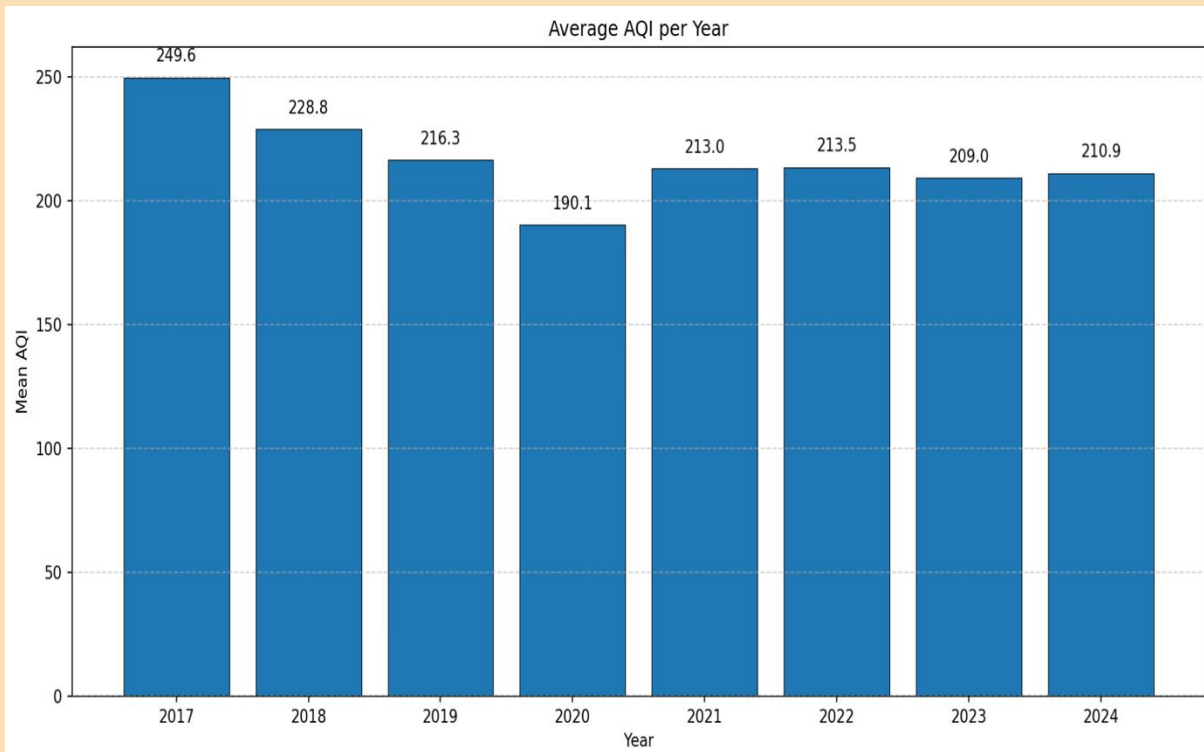
- **Accuracy**
- **Macro-F1**
- **Balanced Accuracy**

Further modelling

- **LSTM** as the single deep learning model for time-series AQI prediction.
- **Evaluation:** 5-fold **grouped cross-validation by station** (avoid leakage across stations) + a **temporal hold-out** (2024).

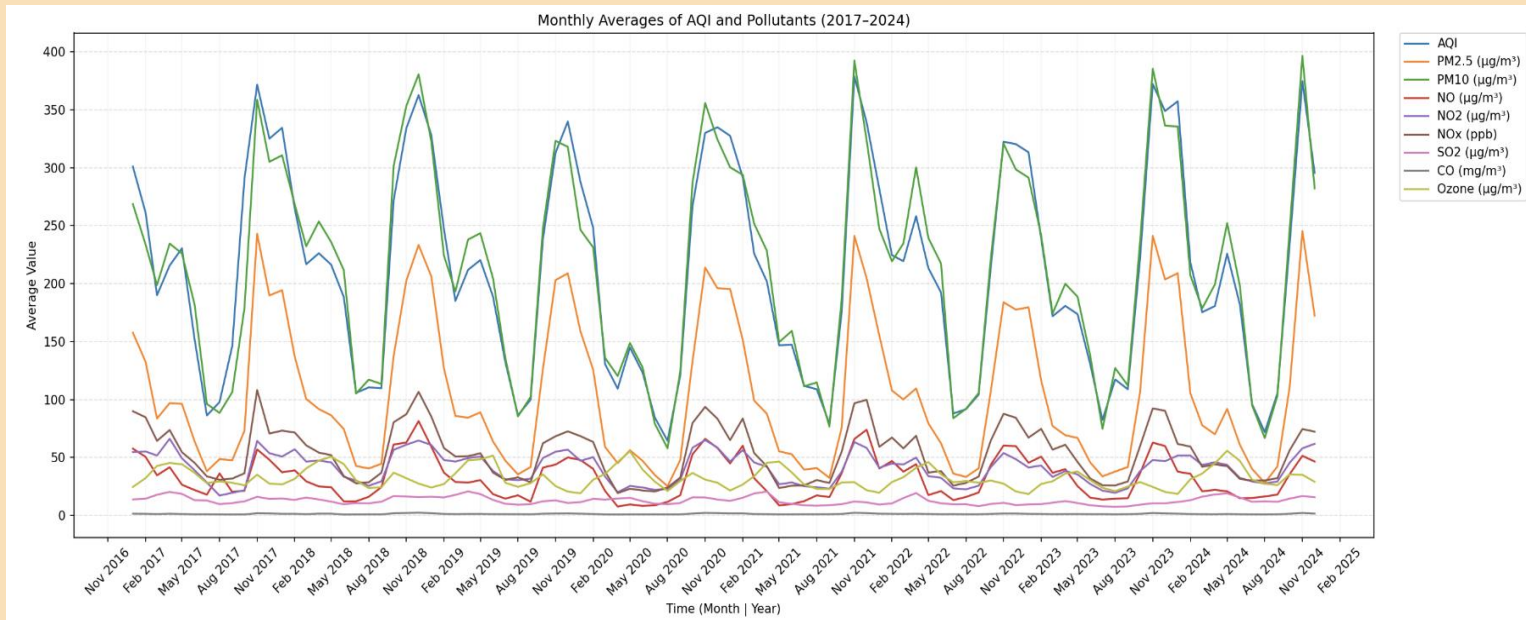
Key Insights

Average AQI per year



- The AQI demonstrates a significant enhancement from 2017 (249.6) to 2020 (190.1), likely attributable to policy measures and the impacts of COVID-19 lockdowns.
- Subsequent to 2020, AQI readings increased once more and maintained near stability, signifying a plateau in advancements and a potential revival of pollution sources.

Monthly Averages of AQI and Pollutants (2017-2024)

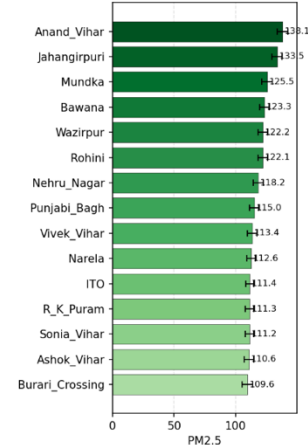


- The AQI, PM2.5, and PM10 exhibit a pronounced seasonal pattern, peaking in the winter months (October to January) and seeing notable declines during the monsoon season (July to August), indicative of seasonal weather effects.
- Winter is also when gaseous pollutants like CO and NOx peak, suggesting that stagnant air conditions increase the concentrations of both particle and gaseous pollutants.

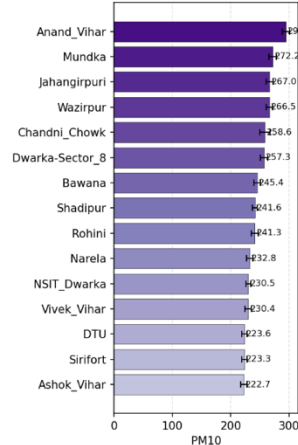
PM2.5, PM10, CO, NO, NO2, SO2 Levels — Top 15 Stations

- Anand Vihar consistently ranks among the top stations for PM2.5, PM10, CO, NO, and NO2, indicating it is a critical hotspot for multiple pollutants.
- Punjabi Bagh has the highest SO2 levels, indicating that SO2 emissions may be more regional in nature and may not always correspond with PM-based pollution trends.

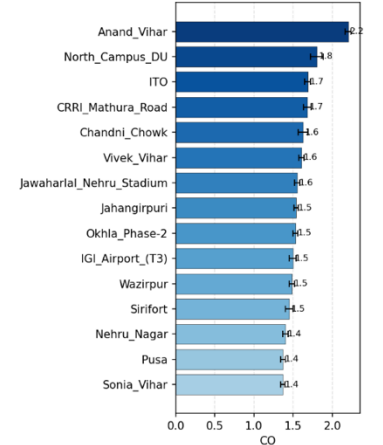
PM2.5 ($\mu\text{g}/\text{m}^3$) Levels — Top 15 Stations



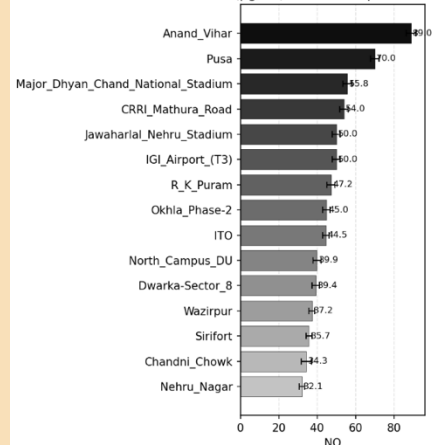
PM10 ($\mu\text{g}/\text{m}^3$) Levels — Top 15 Stations



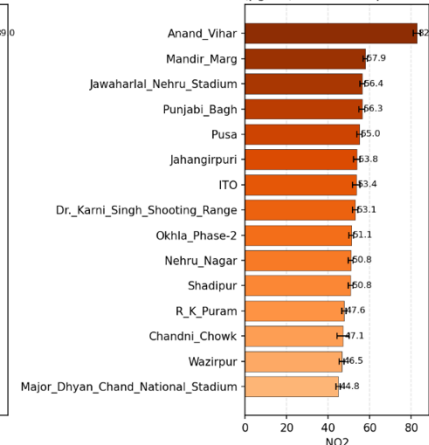
CO (mg/m^3) Levels — Top 15 Stations



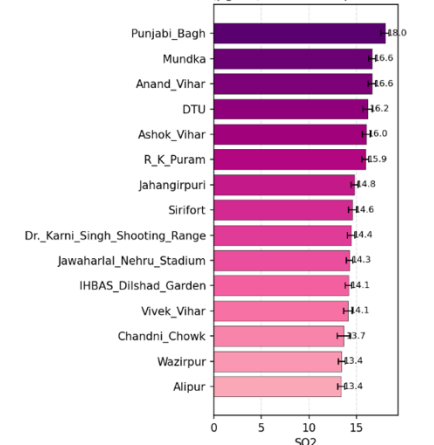
NO ($\mu\text{g}/\text{m}^3$) Levels — Top 15 Stations



NO2 ($\mu\text{g}/\text{m}^3$) Levels — Top 15 Stations



SO2 ($\mu\text{g}/\text{m}^3$) Levels — Top 15 Stations



Expected Outcomes

- A **station-level AQI prediction framework** that incorporates both ground and satellite data.
- Comparative analysis of traditional ML vs LSTM performance.
- Insights into the seasonal and vegetation effects on air quality.

Next Steps

- **AOD (MODIS MAIAC) 6-month means** → compare with AQI, PM2.5 and PM10.
- Train & tune the **6 regressors + 3 classifiers**; compare metrics on **raw vs log** transforms.
- **Explainability**: Permutation significance and SHAP for tree models; partial reliance for PM and NDVI.
- **Stress tests**: Seasonal models and station-specific error analysis.
- Prepare the dissertation.

Conclusion

- This research aims to bridge the gap between ground-based pollution monitoring and satellite-derived environmental indicators for improved AQI prediction in Delhi.
- By combining multiple data sources and modelling techniques, the project will establish robust baselines and highlight the role of vegetation in air quality management.

Thank You

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Link to the video: [COMM514_740064278.mp4](#)